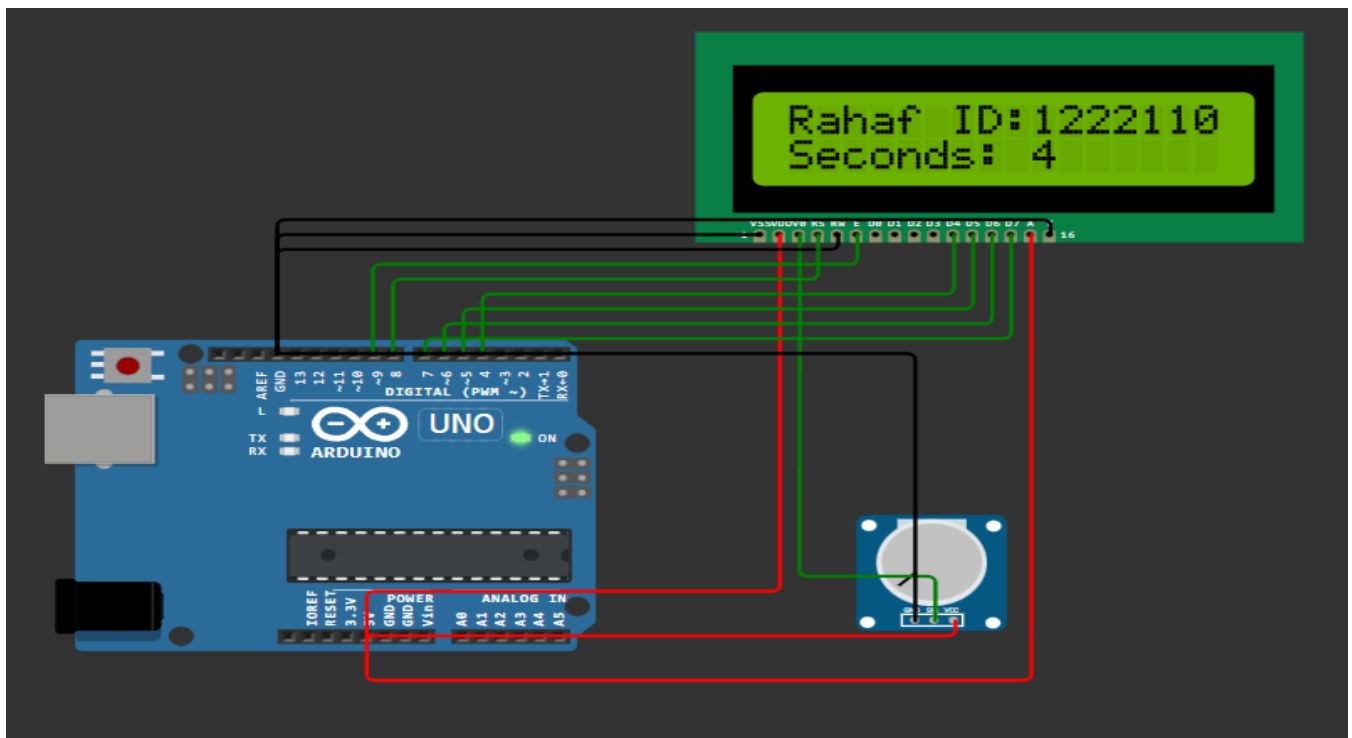


Question 1 — Bare-Metal HD44780 LCD Driver (no library, port masking)



This project uses an Arduino Uno connected to a 16x2 LCD in 4-bit mode to display information on the screen. The first line of the LCD shows the student name and ID, while the second line displays a counter that increases every second. The LCD RS pin is connected to Arduino pin 8 and the Enable pin is connected to pin 9. The LCD data pins D4, D5, D6, and D7 are connected to Arduino digital pins 4, 5, 6, and 7. The VSS and K pins are connected to GND, while VDD and A are connected to 5V to power the LCD and its backlight. A potentiometer is also connected between 5V and GND, and its signal pin is connected to the V0 pin of the LCD to adjust the display contrast. In the code, the LCD is controlled directly using PORTB and PORTD registers instead of using the LCD library. The program sends commands and characters to the LCD, initializes it in 4-bit mode, and uses the millis() function to count elapsed seconds without stopping the program execution.